

MACHINE LEARNING FOR SIGNAL PROCESSING

20-1-2025

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DECISION THEORY (PRML CHAP. 1.5)

❖ Decision Theory

✓ Inference problem

- ◉ Finding the joint density

✓ Decision problem $p(\mathbf{x}, \mathbf{t})$

- ◉ Using the inference to make the classification or regression decision

DECISION PROBLEM - CLASSIFICATION

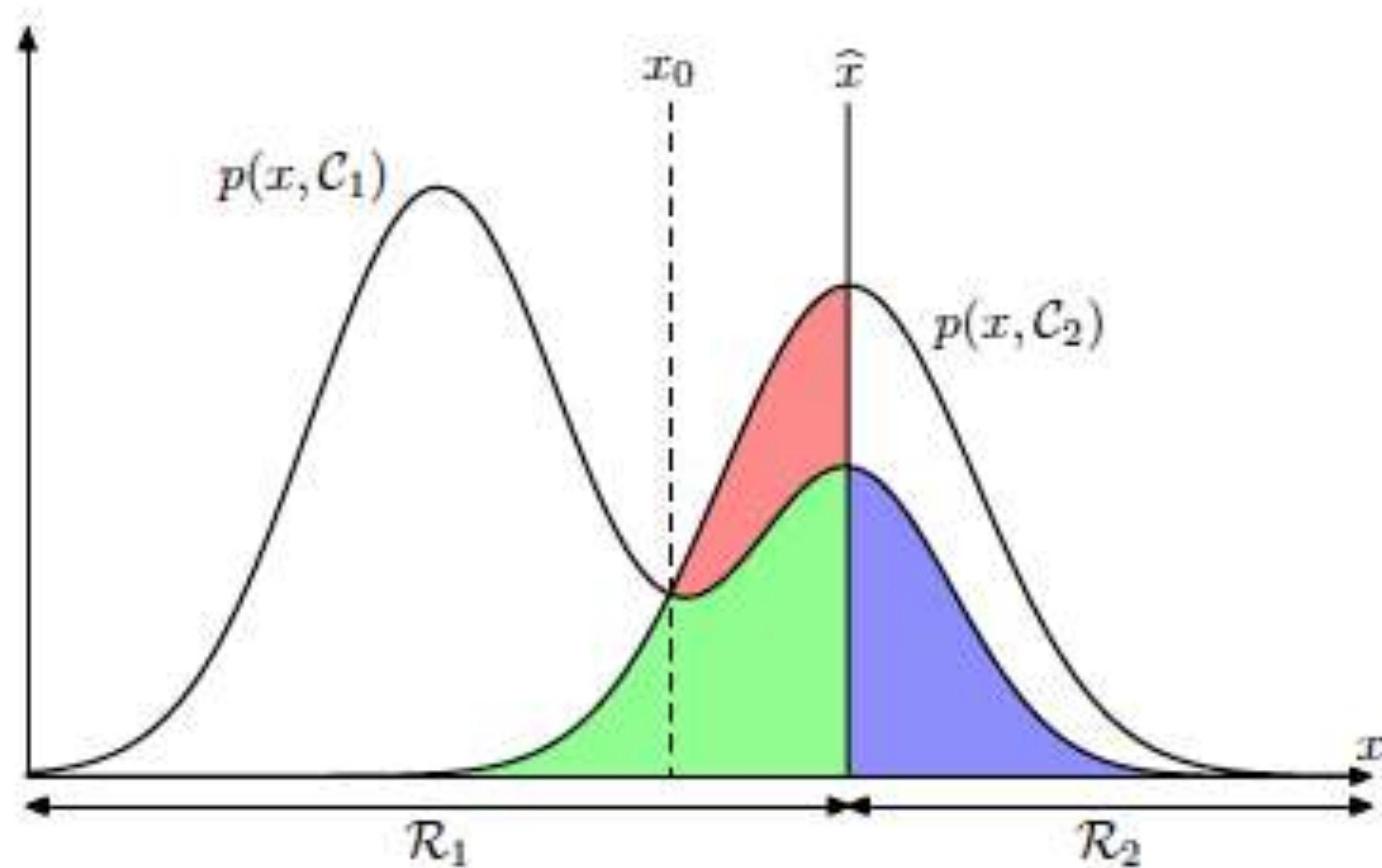
- ✓ Minimizing the mis-classification error
- ✓ Decision based on maximum posteriors

$$\operatorname{argmax}_j p(C_j|\mathbf{x})$$

- ✓ Loss matrix
 - ◉ Minimizing the expected loss

$$\operatorname{argmax}_j \sum_k L_{k,j} p(C_k|\mathbf{x})$$

VISUALIZING THE MAX. POSTERIOR CLASSIFIER



APPROACHES FOR INFERENCE AND DECISION

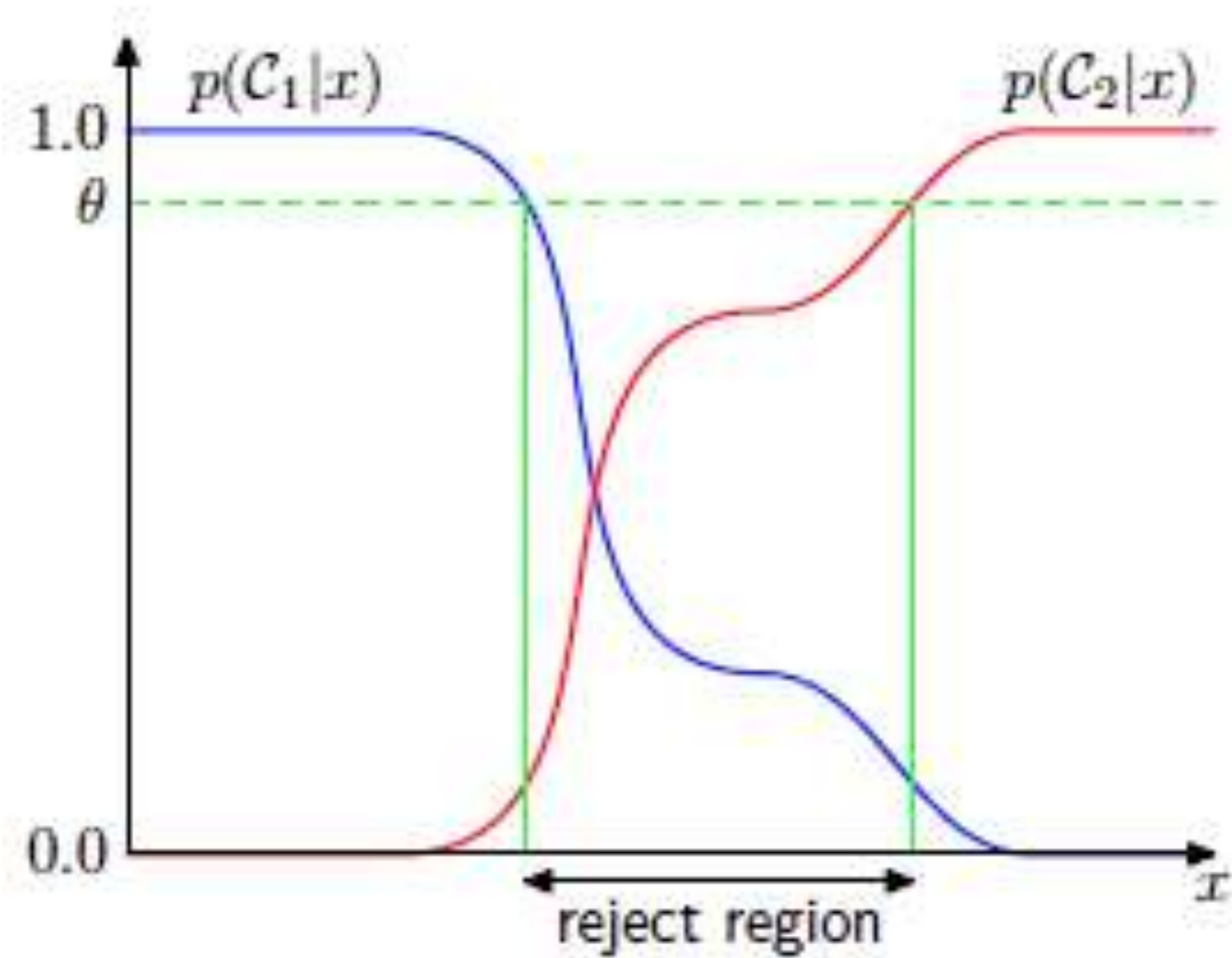
I. Finding the joint density from the data.

II. Finding the posteriors directly.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

III. Using discriminant functions for classification.

ADVANTAGE OF POSTERIORIORS

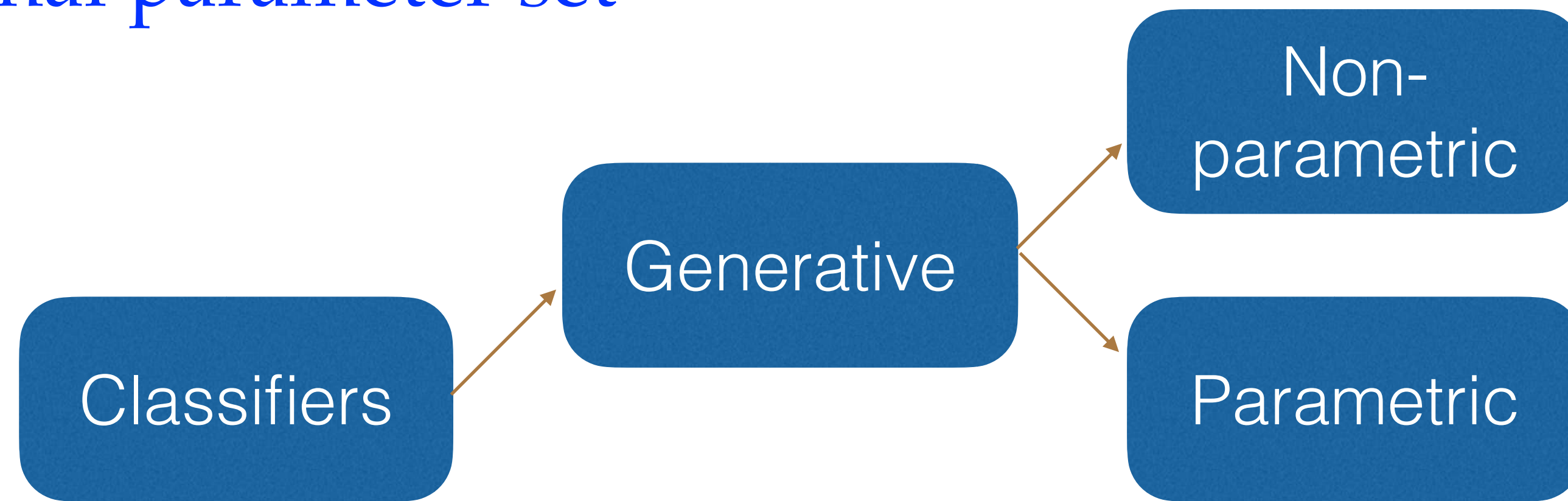


Decision Rule for Regression

- ❖ Minimum mean square error loss
- ❖ Solution is conditional expectation.

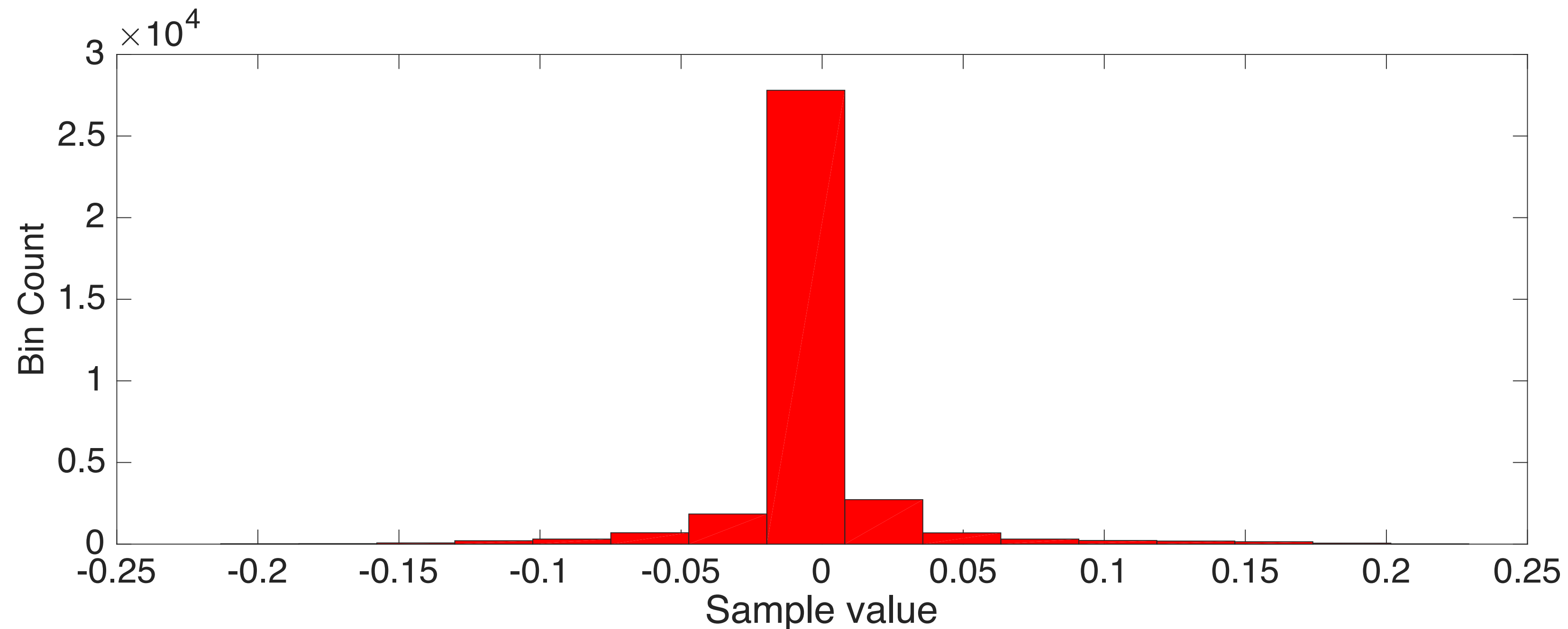
GENERATIVE MODELING

- ❖ Collection of probability distributions which are described by a **finite dimensional parameter set**



Non-parametric Modeling

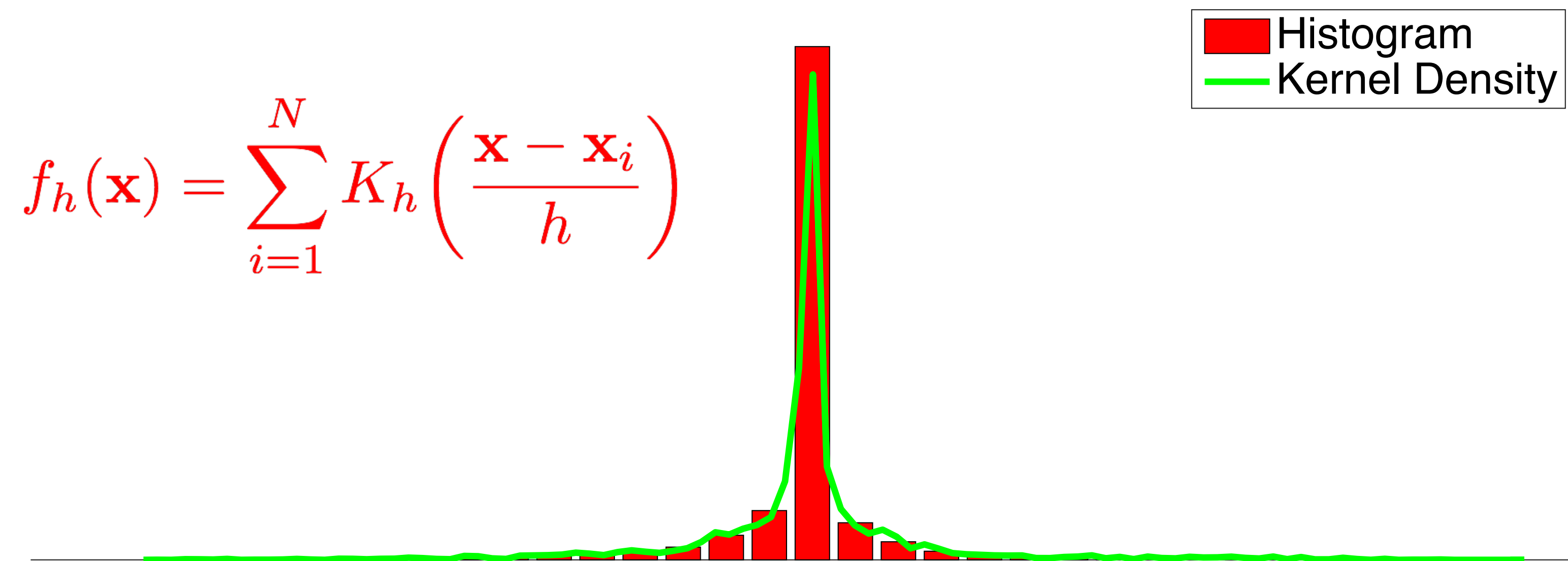
- **Non-parametric** models do not specify an apriori set of parameters to model the distribution. Example - Histogram



The density is not smooth and has block like shape.

Non-parametric Modeling

- **Non-parametric** models do not specify an apriori set of parameters to model the distribution.
- Example - Kernel Density Estimators



Kernel is a smooth function which obeys certain properties

Non-parametric Modeling

- Non-parametric methods are dependent on number of data points
 - Estimation is difficult for large datasets.
- Likelihood computation and model comparisons are hard.
- Limited use in classifiers

Parametric Models (Chap 2 PRML)

- ❖ Collection of probability distributions which are described by a **finite dimensional parameter set**

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K) \quad P = \{P_{\boldsymbol{\theta}}\}$$

- Examples -

- Poisson Distribution

$$p_{\lambda}(j) = \frac{\lambda^j}{j!} e^{-\lambda}$$

- Bernoulli Distribution

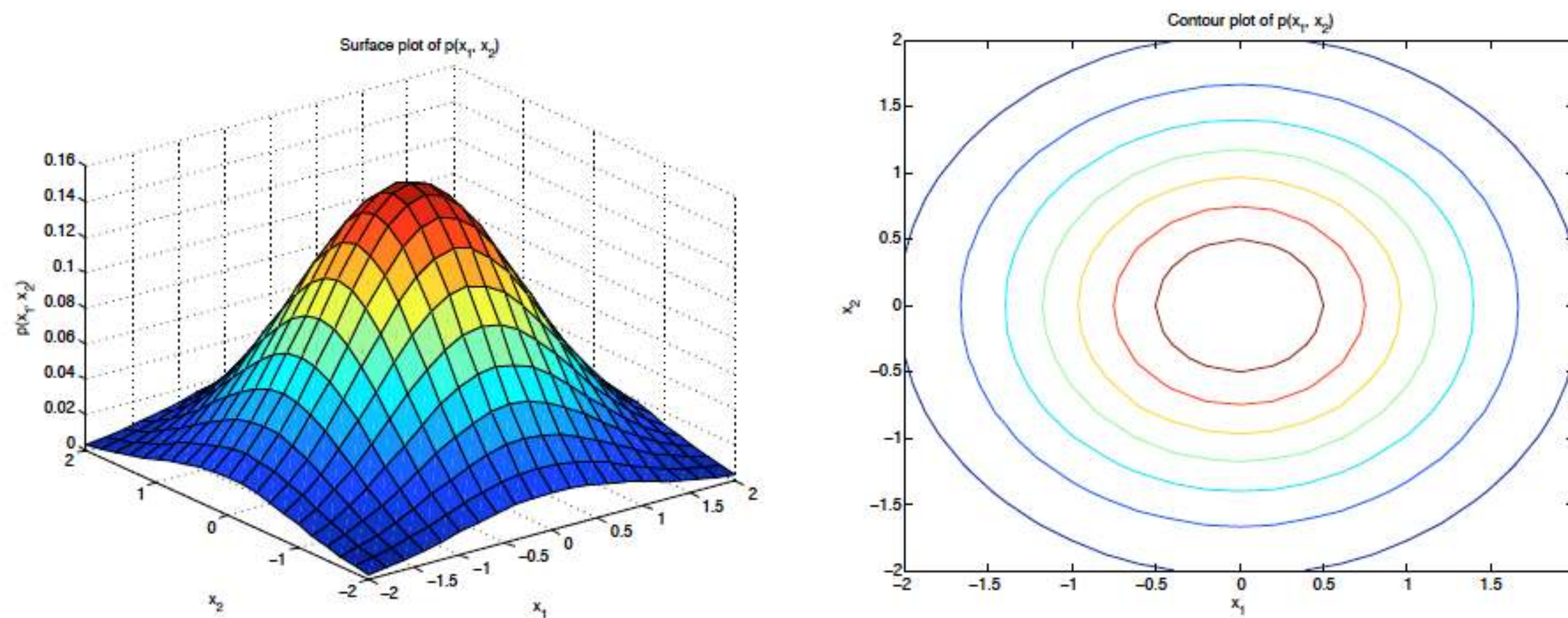
$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{i=1}^D \mu_i^{x_i} (1 - \mu_i)^{1-x_i}$$

- Gaussian Distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Gaussian Distribution

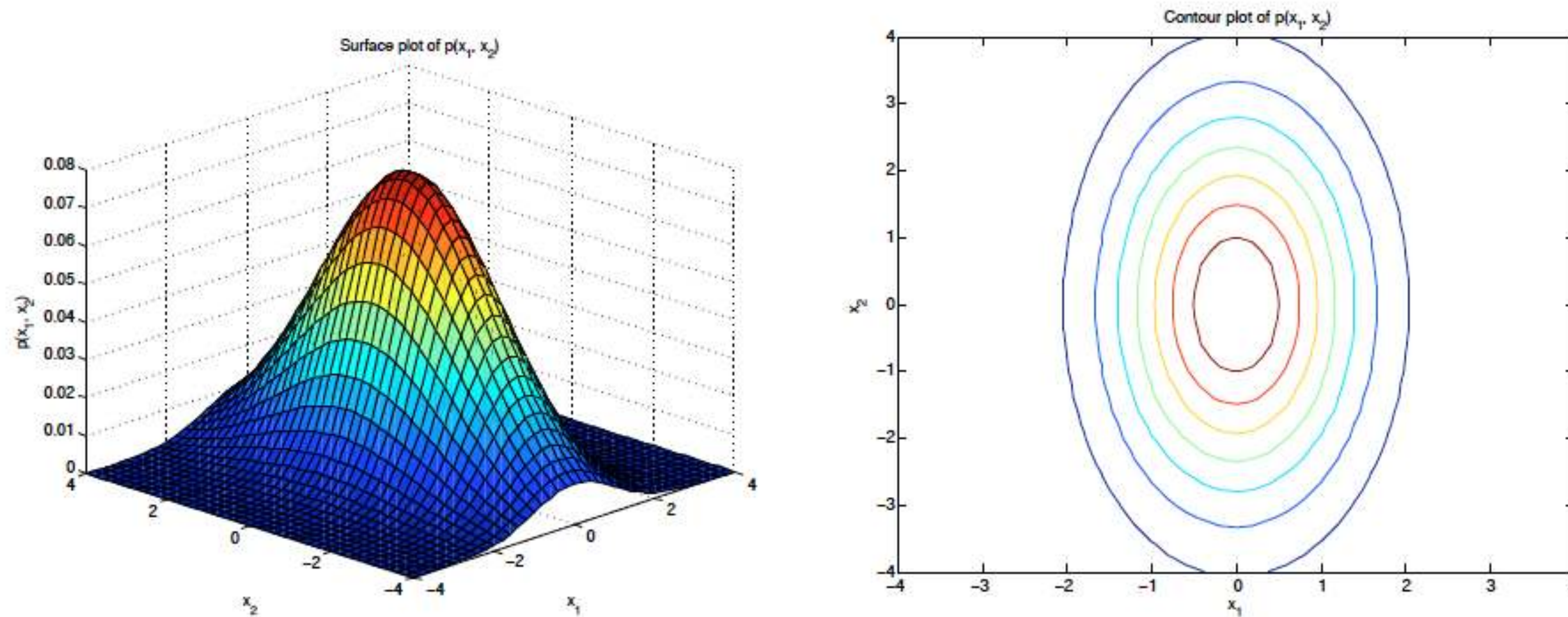
One of most widely used and well studied model



Points of equal probability lie on on contour
Diagonal Gaussian with Identical Variance

Gaussian Distribution

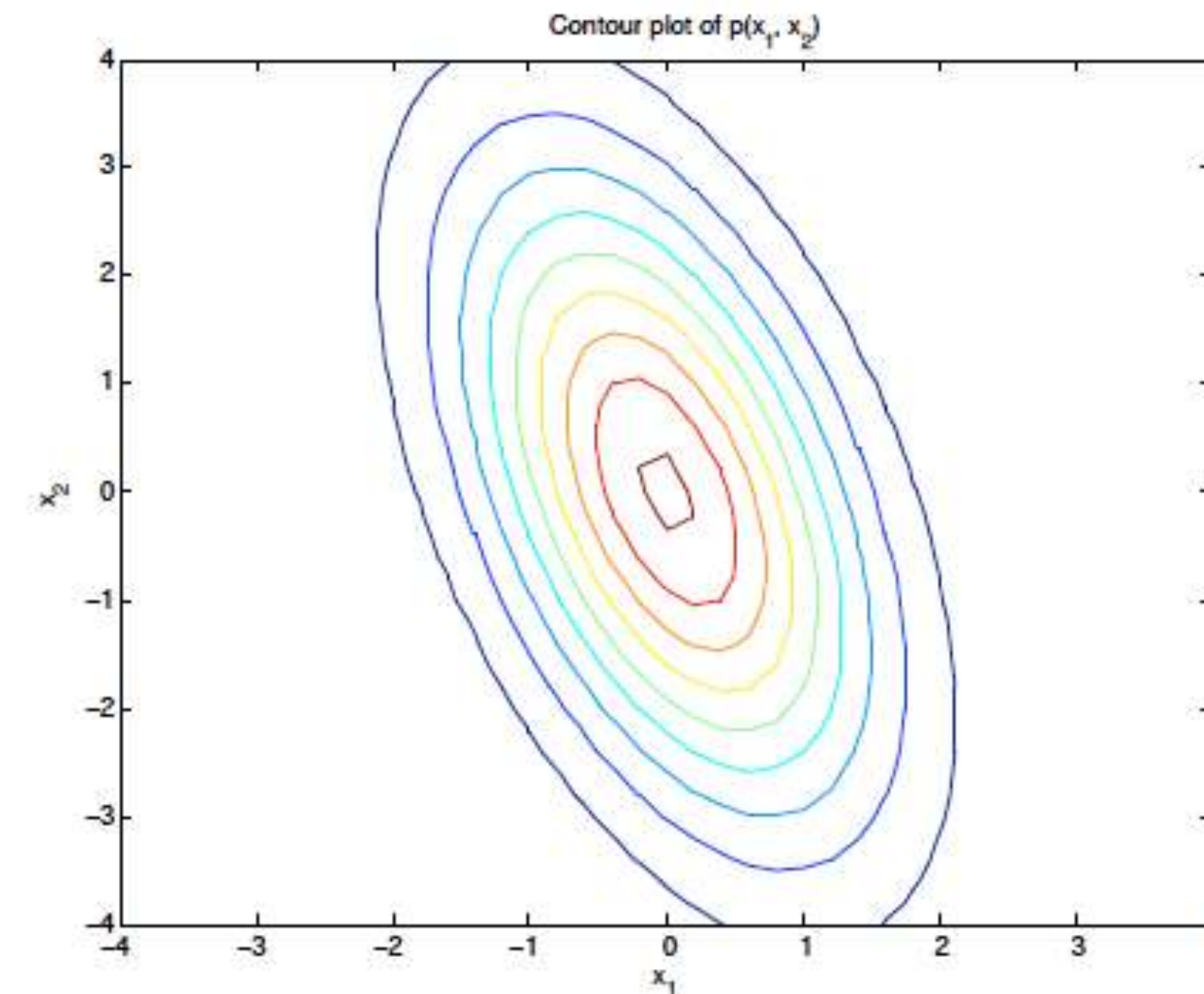
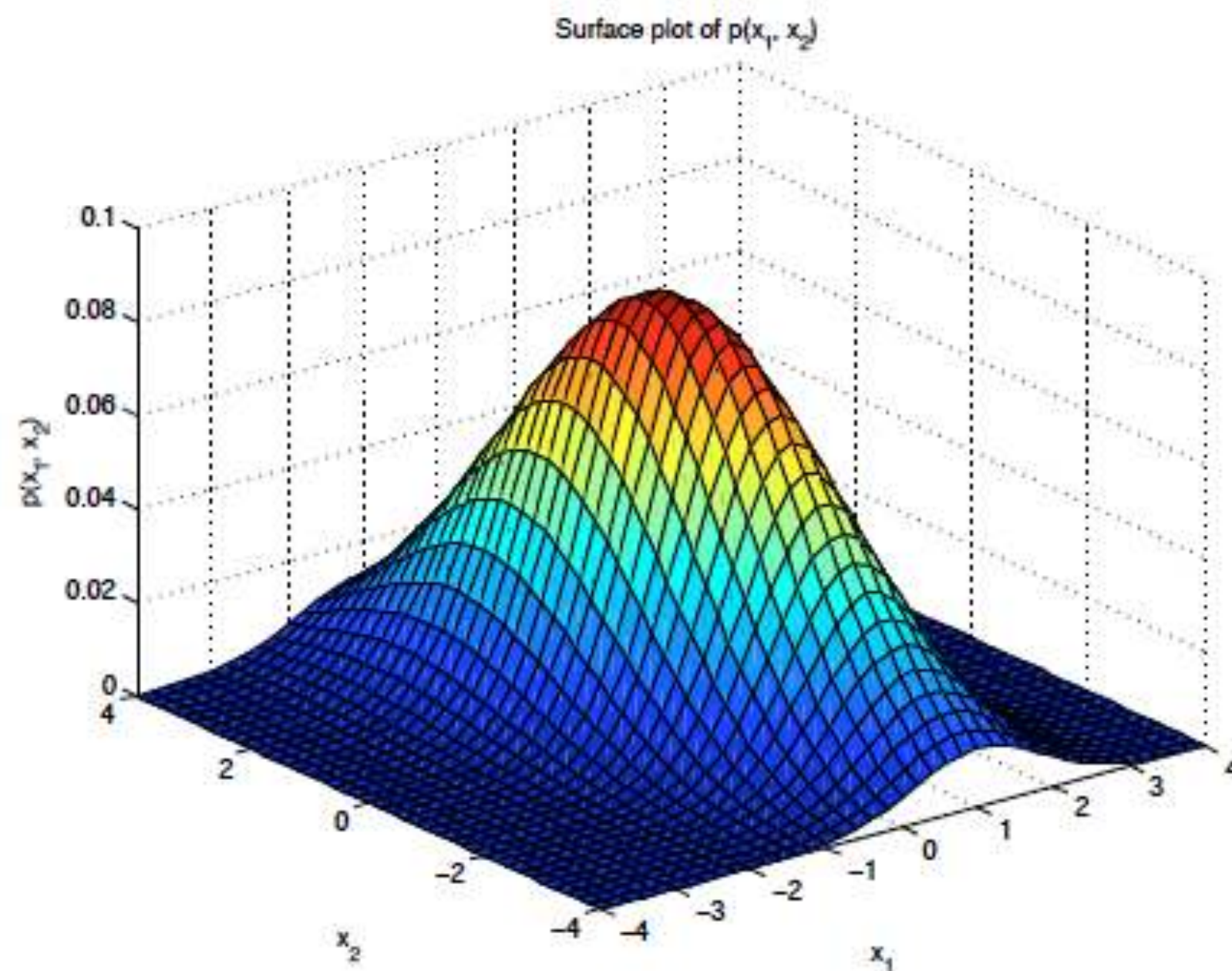
Insights into two dimensional Gaussian distribution



Diagonal Gaussian with different variance

Gaussian Distribution

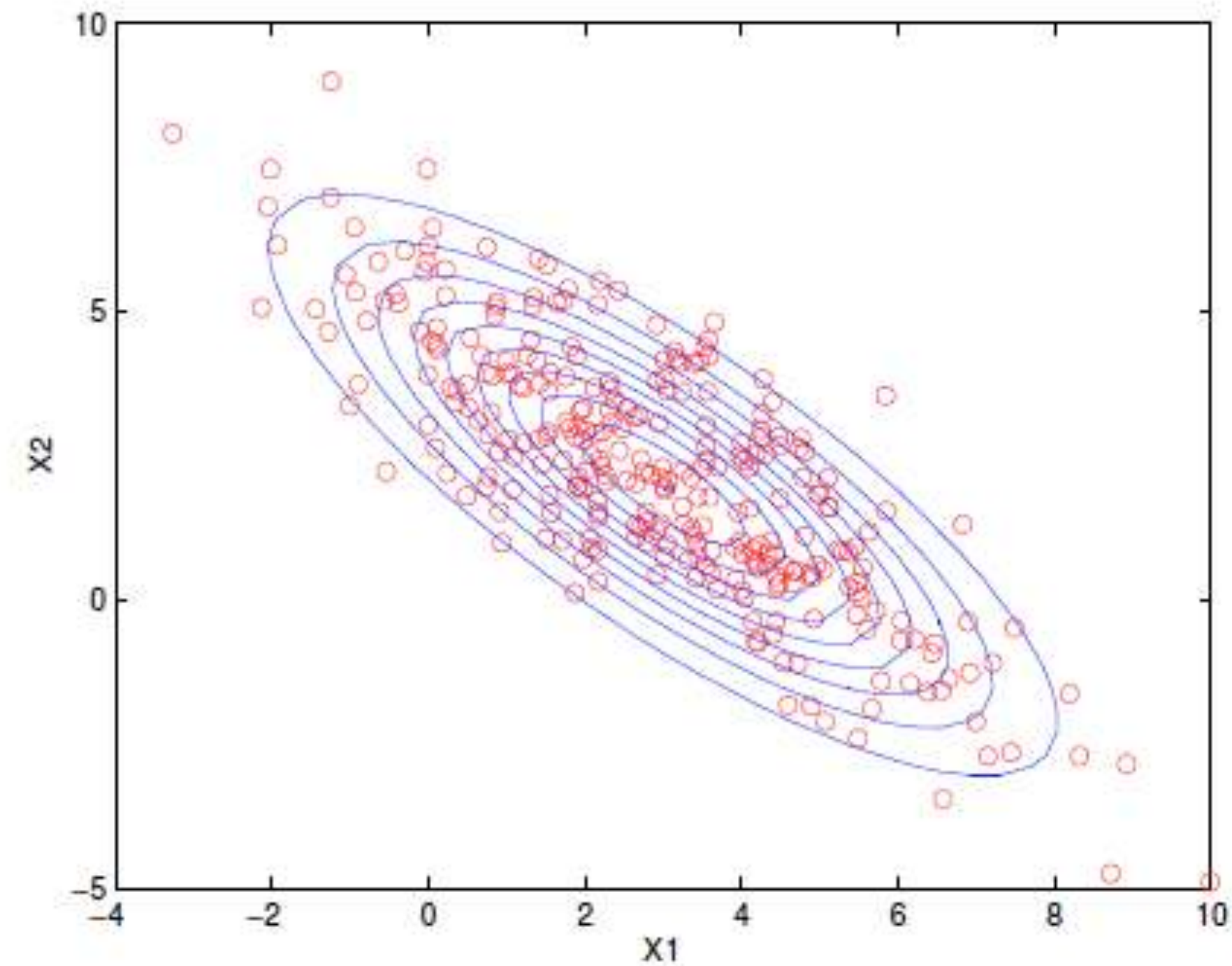
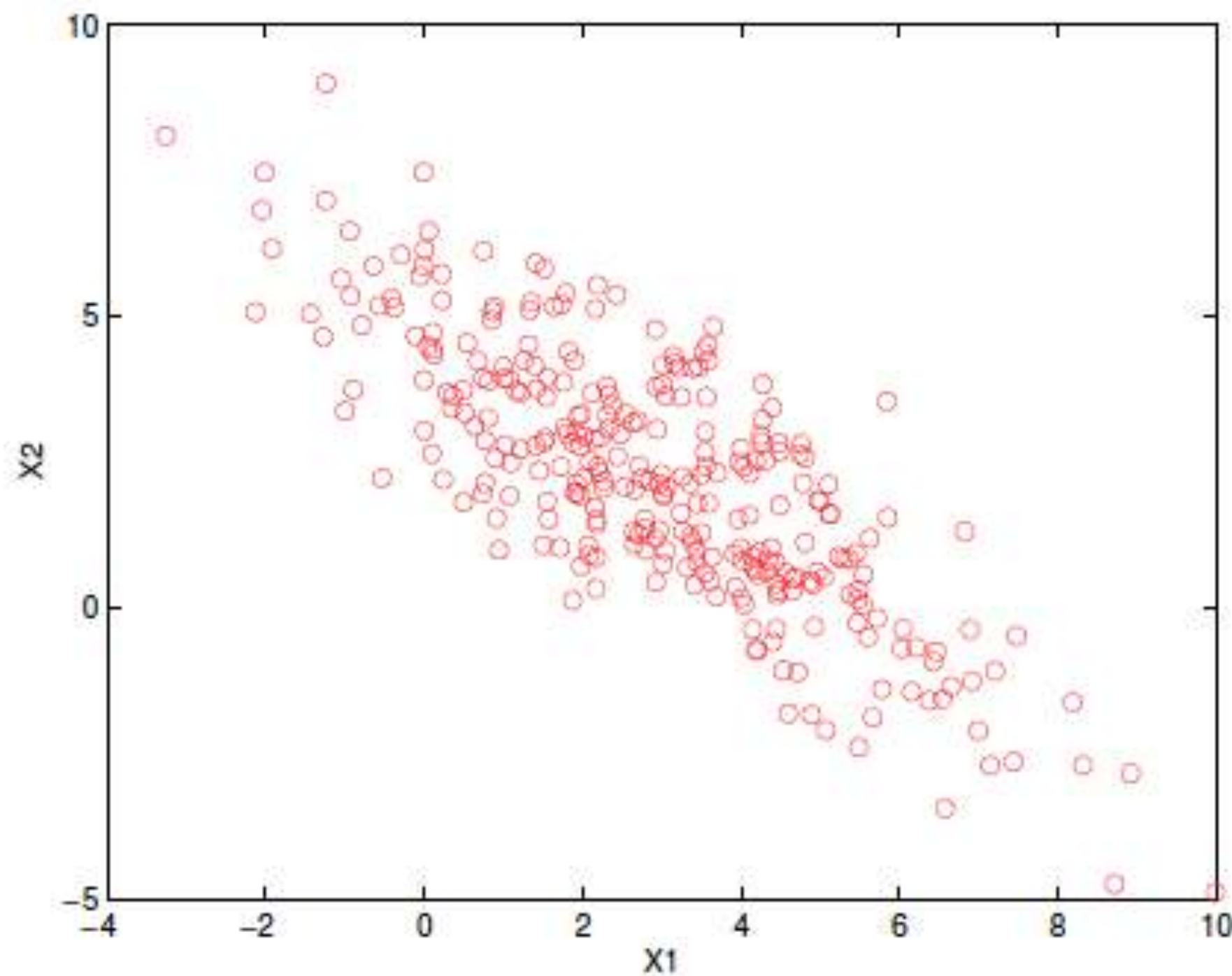
Insights into two dimensional Gaussian Distribution



Full covariance Gaussian distribution

Gaussian Distribution

Fitting the data with a Gaussian Model



Finding the parameters of the Model

- ✓ The Gaussian model has the following parameters

$$\theta = (\mu, \Sigma)$$

- ✓ Total number of parameters to be learned for D dimensional data is $D^2 + D$
- ✓ Given N data points $\{\mathbf{x}_i\}_{i=1}^N$ how do we estimate the parameters of model.
 - ➔ Several criteria can be used
 - ➔ The most popular method is the maximum likelihood estimation (MLE).

MLE

Define the likelihood function as $L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i | \boldsymbol{\theta})$

The maximum likelihood estimator (MLE) is

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{arg\,max}} L(\boldsymbol{\theta})$$

The MLE satisfies nice properties like

- Consistency (convergence to true value)
- Efficiency (has the least Mean squared error).

MLE

For the Gaussian distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp\left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$

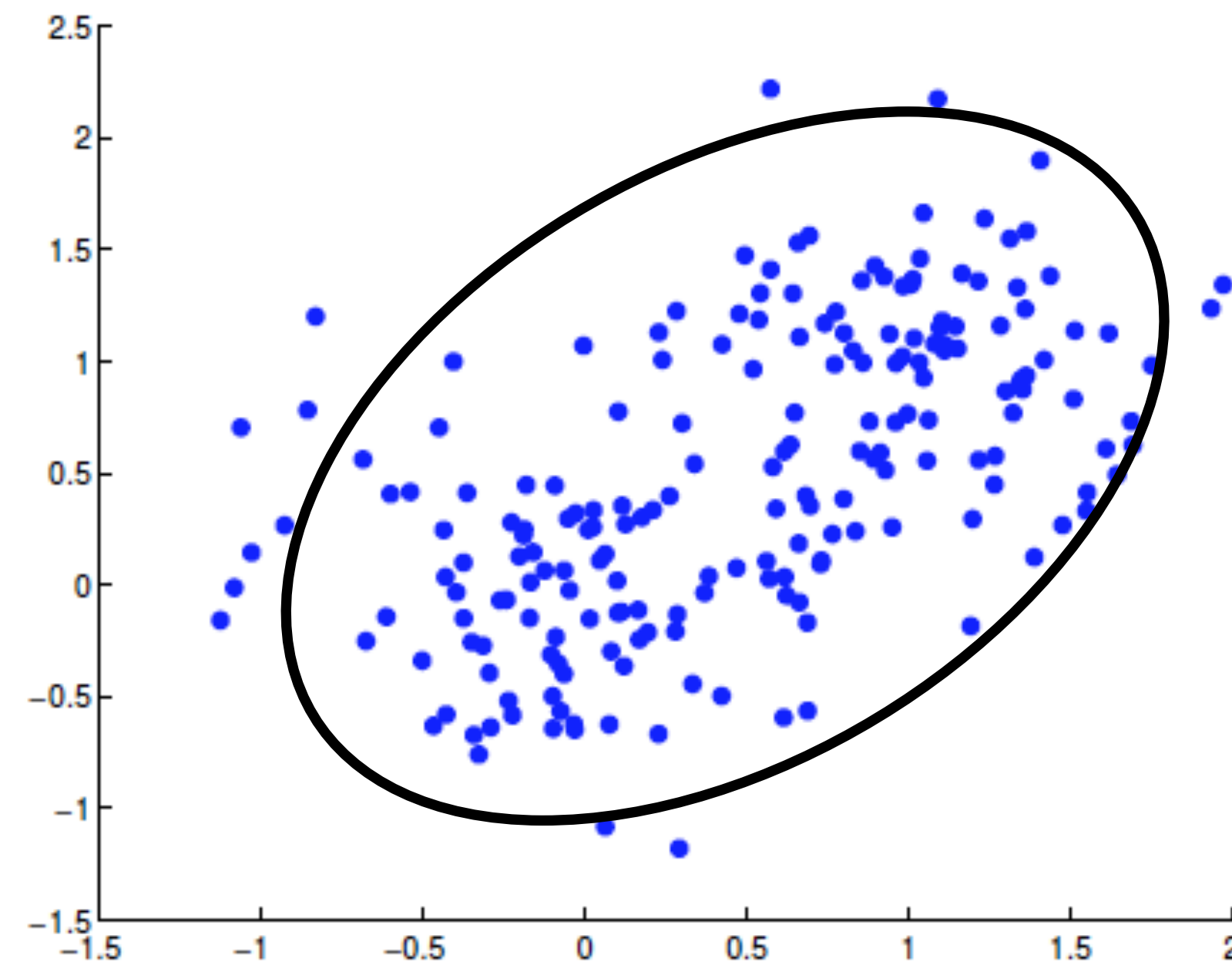
$$L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i|\boldsymbol{\theta})$$

$$\log L(\boldsymbol{\theta}) = -\frac{ND}{2} - \frac{N}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{i=1}^N \left((\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right)$$

To estimate the parameters $\frac{\partial \log L}{\partial \boldsymbol{\mu}} = 0$

Gaussian Distribution

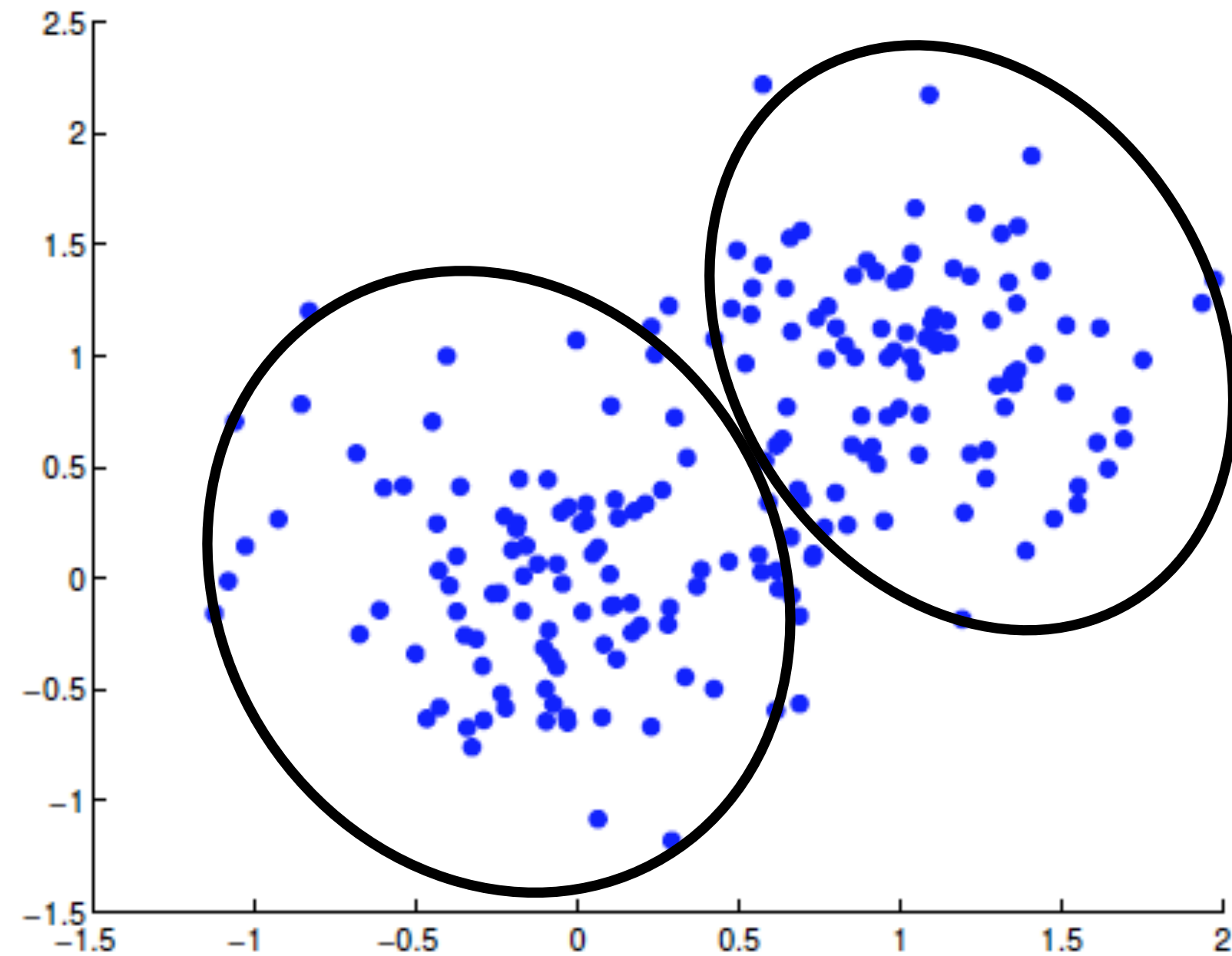
Often the data lies in clusters (2-D example)



Fitting a single Gaussian model may be **too broad**.

Gaussian Distribution

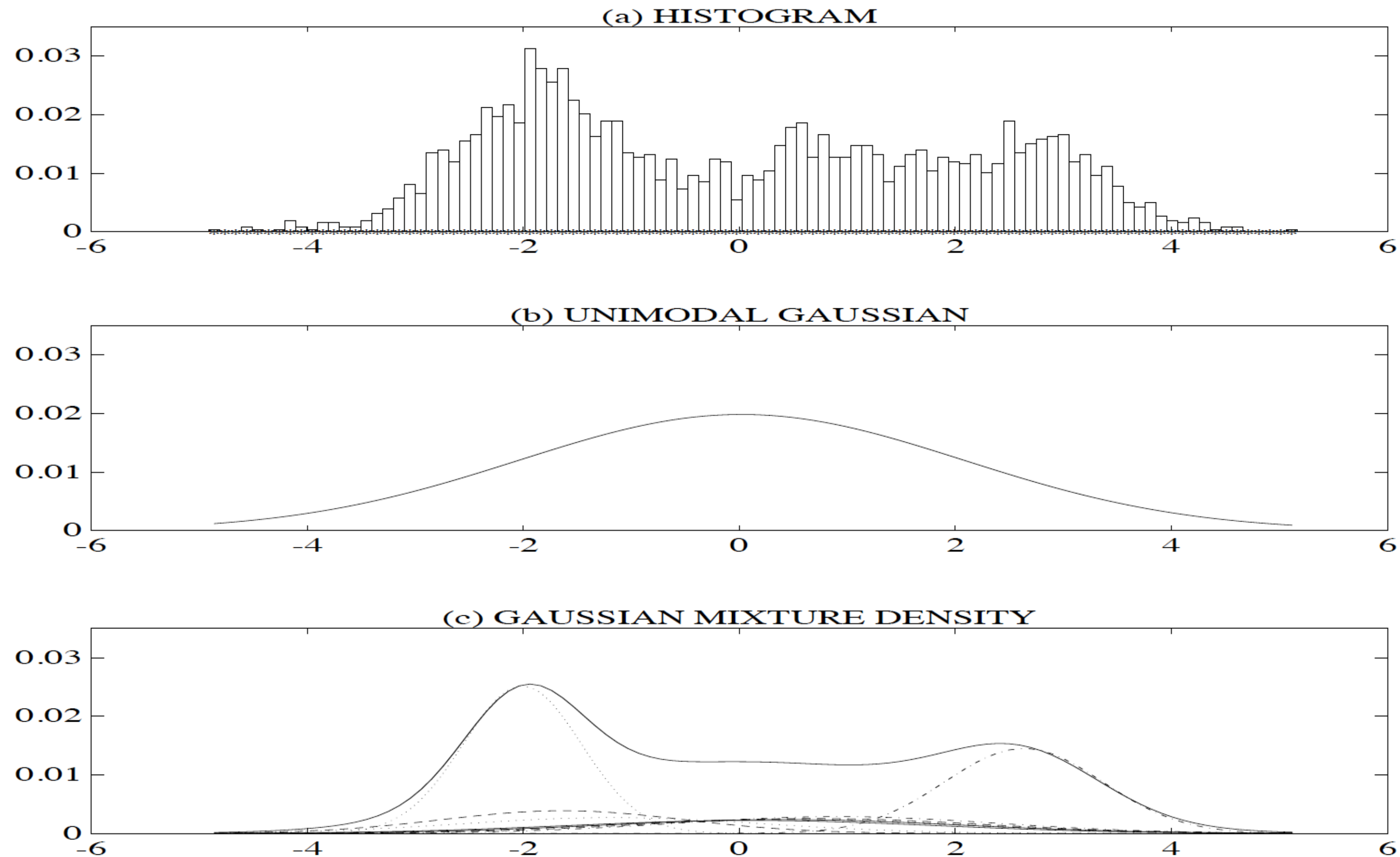
Need mixture models



Can fit any arbitrary distribution.

Gaussian Distribution

Often the data lies in clusters 1-D example



Gaussian Distribution Summary

- ✓ The Gaussian model - parametric distributions
- ✓ Simple and useful properties.
- ✓ Can model unimodal (single peak distributions)
- ✓ MLE gives intuitive results
- ✓ Issues with Gaussian model
 - Multi-modal data
 - Not useful for complex data distributions

Need for mixture models

Gaussian Mixture Models

A Gaussian Mixture Model (GMM) is defined as

$$p(\mathbf{x}|\Theta) = \sum_{k=1}^K \alpha_k p(\mathbf{x}|\theta_k)$$
$$p(\mathbf{x}|\theta_k) = \frac{1}{\sqrt{(2\pi)^D |\Sigma_k|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \mu_k)^* \Sigma_k^{-1} (\mathbf{x} - \mu_k) \right\}$$

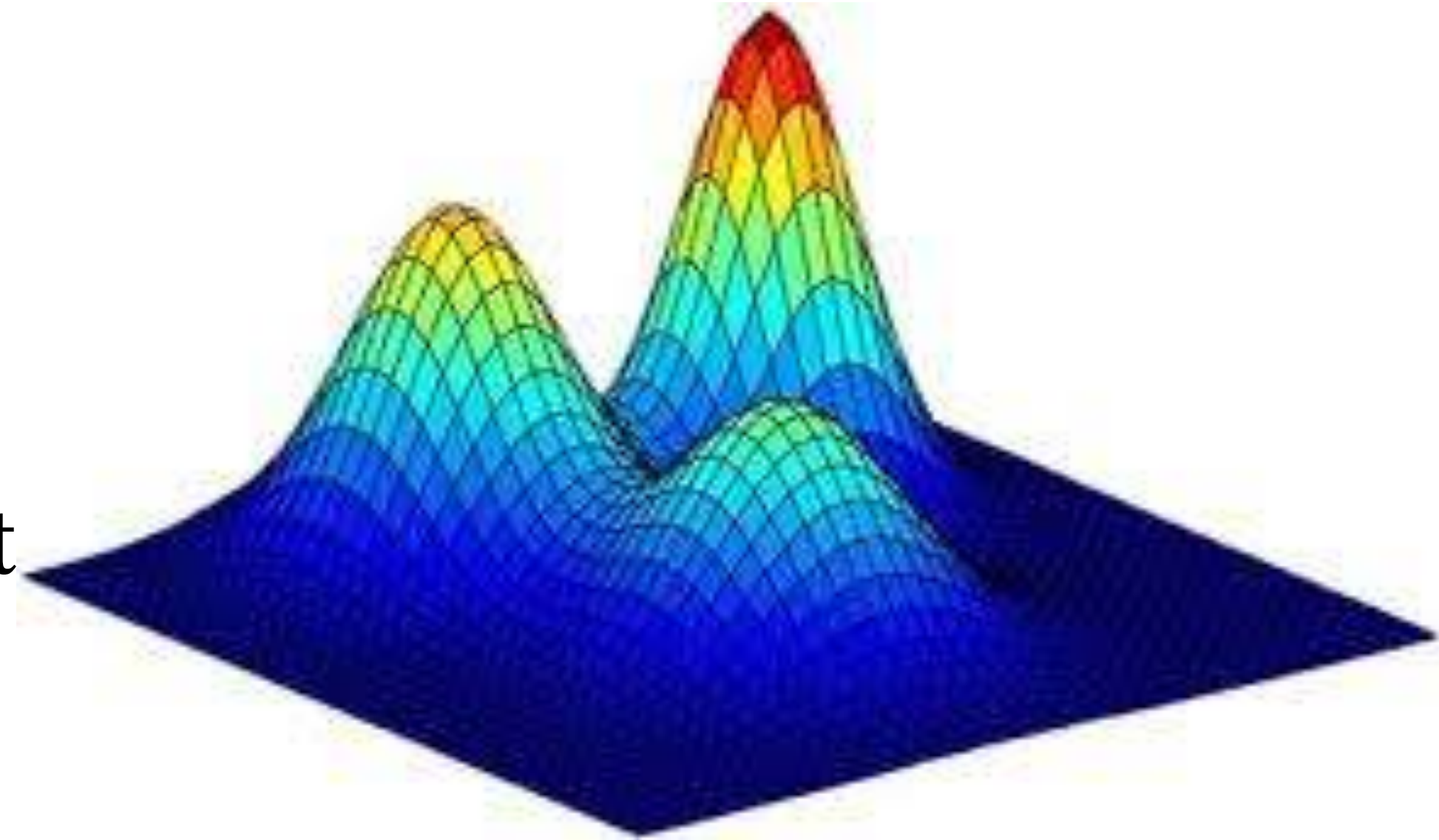
The weighting coefficients have the property

$$\sum_{k=1}^K \alpha_k = 1$$

Gaussian Mixture Models

❖ Properties of GMM

- ✓ Can model multi-modal data.
- ✓ Identify data clusters.
- ✓ Can model arbitrarily complex data dist



The set of parameters for the model are

$$\Theta_k = \{\alpha_k, \theta_k\}_{k=1}^K \quad \theta_k = \{\mu_k, \Sigma_k\}$$

The number of parameters is $KD^2 + KD + K$

MLE for GMM

- ❖ The log-likelihood function over the entire data in this case will have a **logarithm of a summation**

$$\log L(\Theta) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \alpha_k p(\mathbf{x}_i | \theta_k) \right)$$

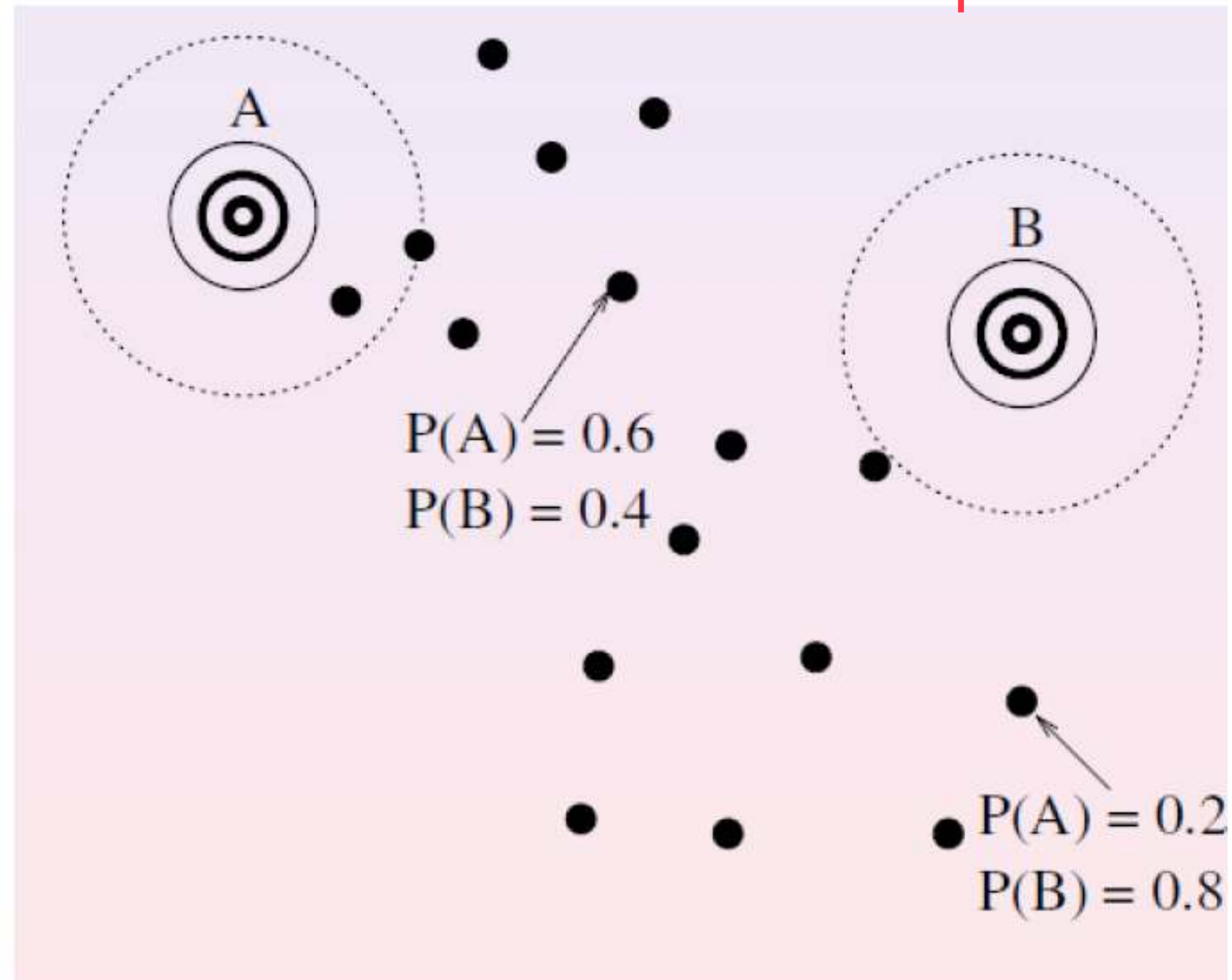
- ❖ Solving for the optimal parameters using MLE for GMM is not straight forward.
- ❖ Resort to the **Expectation Maximization (EM)** algorithm

EM Algorithm For GMMs

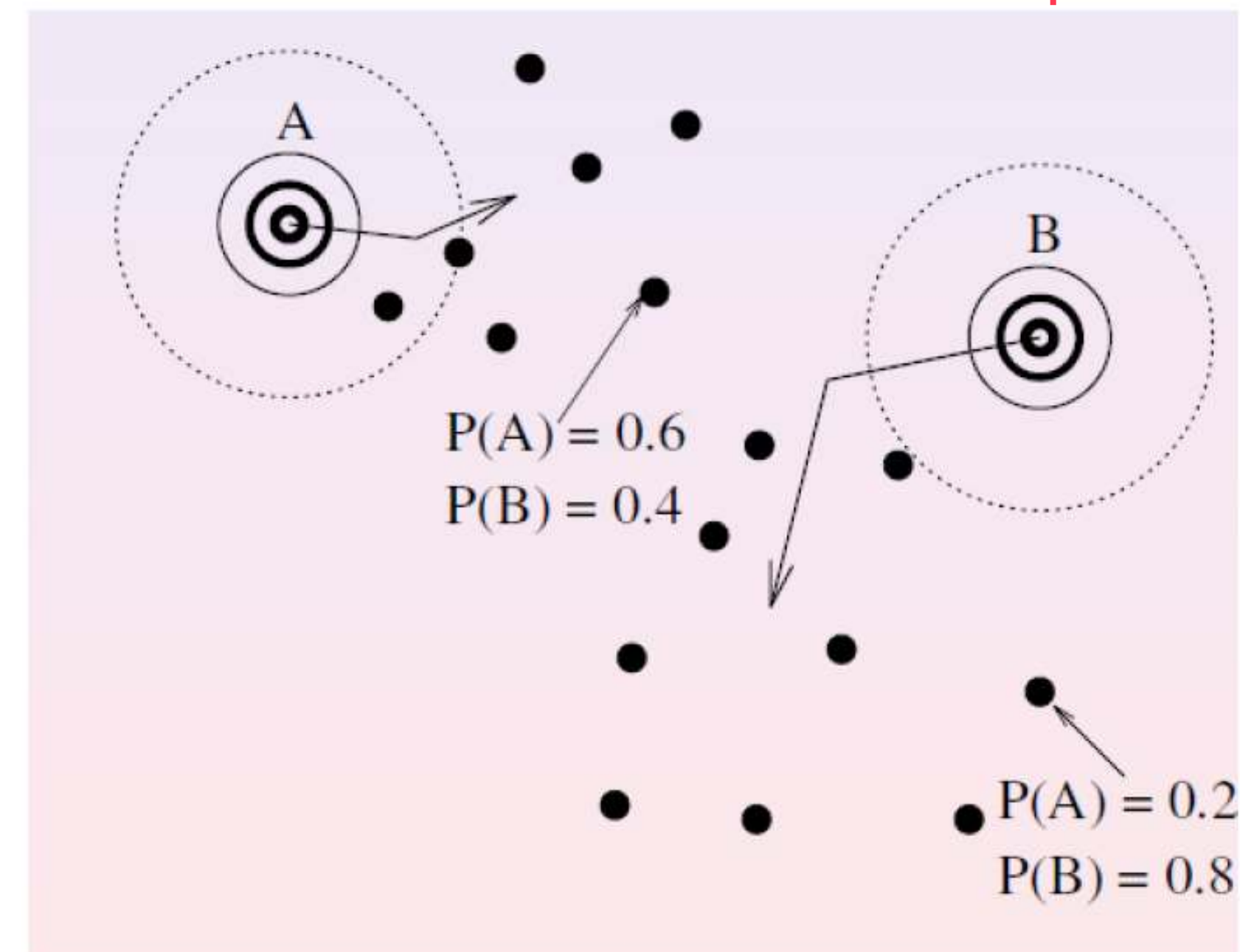
EM Algorithm for GMM

- ✓ The hidden variables will be the index of the mixture component which generated
- ✓ Re-estimation formulae

E-step



M-step



THANK YOU

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